ENTERPRISE BUSINESS & TECHNICAL PROPOSAL

GAGMA Platform

Securing Mobile Banking against Fraudulent APKs

A highly scalable, Graph-Augmented Generative AI solution designed specifically for commercial banking ecosystems. GAGMA automates threat intelligence, reverse-engineers rogue APKs at machine speed, maps structural relationship trees, and proactively isolates digital payment and credential-stealing banking trojans.

PREPARED BY:

GAGMA Engineering & Threat Intelligence Group

Target Audience: Chief Information Security Officer (CISO) & Head of Fraud Operations

Technical Standards: RBI Cyber Security Master Circular Aligned

Deployment Architecture: Hybrid Cloud / On-Premise Sandbox API

1. Executive Pitch & The Banking Value Case

Rogue Android APK files are currently the single most aggressive vector used by fraudsters targeting retail banking customers. Cybercriminals use SMS phishing, fake payment portals, and fraudulent customer service scams to trick users into installing malicious apps. Once installed, these trojans intercept SMS OTPs, overlay fake login sheets on legitimate banking apps, and bypass all multi-factor authentication protocols, executing unauthorized financial transfers.

- Preventative Financial Shield (ROI)

Rather than reacting after fraud has taken place, GAGMA stops the threat at the customer's phone edge. By detecting and blocklisting fraudulent APKs, we stop credential and OTP theft before any transaction occurs, saving millions in customer fraud reimbursements.

- Explainable AI Risk Appraisals

Standard threat scanners flag software as suspicious without context. GAGMA's Graph-Augmented GenAI traces the precise call chains, explaining exactly which permissions, APIs, and domains are used to exfiltrate financial data, satisfying risk compliance requirements.

- Unmatched Operational Velocity

Manual analysis of a complex APK takes up to 48 hours for a skilled security analyst. GAGMA disassembles, maps to Neo4j, runs multi-agent code analysis, and produces CERT-In compliant reports in less than 60 seconds.

2. Technical Architecture & Tech Stack

GAGMA replaces outdated binary scanners with a modern, database-relational approach to code structure. By parsing variables and calls into an interconnected Neo4j graph database, we create a highly queryable knowledge representation of the threat.

Layer / Component	*Technologies Used*	Business & Operational Function
Static Analysis Engine	*jadx, apktool, androguard*	Headless disassembly of compiled DEX files to recover source structu
Graph Knowledge Base	*Neo4j Enterprise / AuraDB*	Dynamic storage mapping class sequences and relationship paths.
GenAI Reasoning Core	*Google Gemini Pro / GPT APIs*	Autonomous code audit, de-obfuscation, and report translation.
API & Integration Gateway	*FastAPI, SQLite3, Caddy Server*	High-performance endpoint routers with auto-SSL routing.
CSO Response Interface	*HTML5, Vanilla CSS, Vis.js*	Glassmorphic dashboard showing call-graphs and phone overlays.

3. Regulatory Compliance & RBI Security Standards

Digital safety mandates set by the Reserve Bank of India (RBI) require banks to take active measures to counter rogue applications and exfiltration attempts:

RBI Cybersecurity Circular (2023)

GAGMA implements continuous automated threat scoring for newly emerged overlays, actively protecting APIs and credentials before release.

CERT-In SLA Framework

Under current mandates, banks must disclose severe cyber incidents within 6 hours. GAGMA generates a formatted CERT-In incident log instantly on completion, saving valuable forensic hours.

NPCI Fraud Risk Management

Enforces targeted protection overlays that specifically detect malicious hooks aiming to hijack UPI, Google Pay, PhonePe, and major commercial banking systems.

4. Commercial Implementation Roadmap

Deploying GAGMA into your bank's ecosystem is structured into three seamless, non-disruptive phases designed to guarantee standard security without interfering with the bank's core banking system (CBS):

Phase 1: Passive Threat-Intelligence Sandbox (Weeks 1 - 2)

Establish a dedicated sandbox API endpoint. Ingest newly discovered APK binaries sent via threat-feeds or customer reports directly into the GAGMA sandbox. Run autonomous decompile and graph modeling without affecting any user-facing code.

Phase 2: Active SIEM & MDM Automated Containment (Weeks 3 - 4)

Wire GAGMA's incident webhook outputs straight to the bank's Security Operations Center SIEM (e.g. Splunk or QRadar) and corporate Mobile Device Management (MDM). If a critical threat is mapped, trigger automated blocklists.

Phase 3: Cross-Channel Customer Alerting (Weeks 5 - 6)

Activate GAGMA's redundant mobile gateway. When a malicious campaign targeting your bank's package name is validated, broadcast automated push alerts and blocklist rules to custom mobile apps, keeping customer devices safe.